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Proposed Multi-Horizon Forecasting Update for the Padma Bridge ML Project

Next-Day, Weekly and Monthly Traffic and Toll Forecasting

Course: CSE 4112: Machine Learning Laboratory

Group: B2

Batch: 2021

September 2026

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1 Introduction

The current Padma Bridge machine-learning system predicts total traffic and toll revenue one day ahead. Although next-day prediction is useful for immediate operational planning, decision-makers may also require forecasts covering longer periods.

This update proposes extending the current system so that it can produce daily predictions for the next day, the next seven days and the next thirty days. The updated system will also calculate total and average traffic and toll revenue for the weekly and monthly forecast periods.

2 Objectives

The objectives of the proposed update are:

- To preserve the existing one-day forecasting system as a baseline.
- To forecast daily Padma Bridge traffic for up to thirty days.
- To forecast daily Padma Bridge toll revenue for up to thirty days.
- To calculate weekly and monthly totals and daily averages.
- To include future weekends, public holidays and Eid periods in the forecasting process.
- To compare multiple machine-learning models at different forecasting horizons.
- To evaluate how forecasting accuracy changes as the prediction horizon increases.

3 Existing Forecasting System

The existing system predicts the value for the following day:

$$\hat{y}_{t+1} = f(X_t), \tag{1}$$

where:

- X_t represents the information available up to day t ;
- $\hat{y}_{t+1}$ is the predicted traffic or toll revenue for the following day.

The current system compares Linear Regression, Random Forest, XGBoost, SARIMAX and a custom Residual Multilayer Perceptron. Random Forest produced the lowest average traffic error, while the Residual MLP produced the lowest average toll-revenue error.

4 Need for Model Retraining

The existing models cannot directly produce reliable seven-day or thirty-day forecasts because they were trained only for one-day-ahead prediction.

The current relationship is:

$$X_t \longrightarrow y_{t+1}. \quad (2)$$

The proposed system must learn:

$$X_t, h \longrightarrow y_{t+h}, \quad h = 1, 2, \dots, 30, \quad (3)$$

where h represents the number of days between the observation date and the target date.

Therefore, the existing models should not simply be reused without training. The feature-engineering and prediction programs must be updated, and the models must be retrained using multiple forecast horizons.

The current one-day models should be retained as baselines for comparison.

5 Proposed Multi-Horizon Method

A global direct multi-horizon forecasting method is recommended. A new input variable named `forecast_horizon` will represent the number of days ahead being predicted.

During training, each historical observation will be expanded into multiple examples.

Table 1: Example construction of multi-horizon training samples

Observation date	Forecast horizon	Target date	Prediction target
1 January	1 day	2 January	Traffic on 2 January
1 January	7 days	8 January	Traffic on 8 January
1 January	30 days	31 January	Traffic on 31 January

The model will learn how the relationship between historical information and the target changes at different forecasting horizons.

6 Input Features

The proposed model will use two groups of inputs.

6.1 Historical Inputs

These values must be available on the day when the forecast is generated:

- Previous Padma Bridge traffic;
- Previous Padma Bridge toll revenue;
- Traffic lags of 1, 2, 3, 7, 14 and 28 days;
- Toll-revenue lags;
- Rolling traffic means and standard deviations;
- Historical revenue per vehicle;
- Historical Jamuna Bridge traffic and toll revenue;
- Recent traffic changes and trends.

6.2 Target-Date Inputs

These variables describe the future date being predicted:

- Forecast horizon from 1 to 30 days;
- Day of the week;
- Month and day of the year;
- Weekend indicator;
- Public-holiday indicator;
- Eid-ul-Fitr or Eid-ul-Adha indicator;
- Signed number of days before or after Eid;
- Railway-operation indicator;
- Expected temperature;
- Expected rainfall;
- Expected humidity;
- Expected wind speed.

Calendar, weekend, holiday and Eid variables can be calculated in advance. Weather forecasts should be used when available. For distant dates where reliable weather forecasts are unavailable, historical seasonal weather averages may be used, but the increased uncertainty must be reported.

Table 2: Proposed model comparison

Model	Purpose
Persistence baseline	Uses the latest available traffic or toll value as the future prediction.
Seasonal baseline	Uses the corresponding value from the previous week.
Random Forest	Preserves the current strongest average-error traffic model and captures nonlinear relationships.
XGBoost	Provides a strong boosted-tree model and is useful for reducing large forecasting errors.
CatBoost	Provides an additional event-aware boosting comparison and can effectively model holiday and calendar variables.
Residual MLP	Provides a neural-network comparison by predicting changes from recent traffic or toll levels.

7 Models to Be Compared

The updated project should compare several forecasting approaches.

Random Forest, XGBoost and CatBoost should be the main models in the first implementation. The Residual MLP can be retained as an additional neural-network comparison.

An LSTM is not essential because the available dataset contains only approximately four years of daily observations, while tree-based models are well suited to small structured datasets with engineered time-series features.

8 Expected Outputs

For each future date, the system will generate:

1. Predicted total Padma Bridge traffic in vehicles;
2. Predicted total Padma Bridge toll revenue in BDT.

8.1 Next-Day Output

Prediction date: 10 September 2026

Predicted traffic: 20,500 vehicles

Predicted toll revenue: BDT 24.8 million

8.2 Seven-Day Output

The weekly forecasting mode will provide:

- Seven daily traffic predictions;
- Seven daily toll-revenue predictions;
- Total weekly traffic;
- Total weekly toll revenue;
- Average daily traffic and toll revenue.

8.3 Thirty-Day Output

The monthly forecasting mode will provide:

- Thirty daily traffic predictions;
- Thirty daily toll-revenue predictions;
- Total traffic for the thirty-day period;
- Total toll revenue for the thirty-day period;
- Average daily traffic and toll revenue.

The output table should also label weekends, public holidays and Eid periods.

9 Model Training Procedure

The following steps will be used to update the system:

1. Preserve the existing cleaned datasets and one-day models.
2. Generate targets for forecast horizons from 1 to 30 days.
3. Add `forecast_horizon` as an input feature.
4. Generate calendar, holiday and Eid features for each target date.
5. Ensure that historical traffic and toll inputs contain only information available on or before the forecast-origin date.
6. Divide the data chronologically into training, validation and test sets.
7. Train Random Forest, XGBoost, CatBoost and Residual MLP models.
8. Tune the models using expanding-window time-series cross-validation.
9. Evaluate every model on the same untouched test period.
10. Select the strongest model for daily, weekly and monthly forecasting.

11. Save the selected models and update the prediction interface.

All samples generated from the same forecast-origin date must remain in the same data partition. This prevents information from one original date from appearing in both the training and test sets.

10 Model Evaluation

The models will be evaluated using:

- Mean Absolute Error (MAE);
- Root Mean Squared Error (RMSE);
- Mean Absolute Percentage Error (MAPE);
- Coefficient of Determination (R^2).

Performance should be reported separately for:

- One-day-ahead forecasts;
- Forecasts from days 2 to 7;
- Forecasts from days 8 to 30;
- Normal days;
- Public holidays;
- Eid periods.

Table 3: Structure of the multi-horizon model comparison

Model	1-day MAPE	7-day MAPE	30-day MAPE	Eid MAPE
Random Forest	—	—	—	—
XGBoost	—	—	—	—
CatBoost	—	—	—	—
Residual MLP	—	—	—	—

Errors are expected to increase with the forecasting horizon:

$$\text{Error}_{1 \text{ day}} < \text{Error}_{7 \text{ days}} < \text{Error}_{30 \text{ days}}. \quad (4)$$

Weekly and monthly total errors should also be calculated by comparing the sum of the predicted daily values with the sum of the actual daily values.

11 Prediction Uncertainty

Longer forecasting horizons contain greater uncertainty because exact future weather, traffic conditions and unexpected events are not known.

A future improvement can produce prediction intervals. For example:

Predicted traffic: 20,500 vehicles

90 percent prediction interval: 17,800--24,300 vehicles

Prediction intervals may be produced using quantile boosting or conformal prediction.

12 Expected Limitations

The proposed system will have the following limitations:

- Forecast accuracy will normally decrease for longer horizons.
- Only approximately four years of bridge data are available.
- The dataset contains relatively few Eid and major-holiday examples.
- Future weather forecasts may be inaccurate or unavailable for the complete thirty-day period.
- Future Jamuna Bridge traffic may need to be forecast separately or excluded from longer-horizon models.
- Unexpected events such as accidents, strikes, toll-rate changes and road closures cannot be predicted from the current dataset.

13 Conclusion

The Padma Bridge forecasting project can be extended from one-day prediction to multi-horizon prediction by updating the dataset construction process and retraining the models. The previous one-day models should be preserved as baselines, while the new system should learn forecasts for horizons ranging from one to thirty days.

Random Forest, XGBoost and CatBoost should be compared as the primary models, with the Residual MLP retained as a neural-network comparison. The final system will produce daily traffic and toll-revenue predictions for the next day, next week and next month, together with weekly and monthly totals.

The main contribution of the updated project can be summarized as follows:

This project extends one-day Padma Bridge forecasting into an Eid- and holiday-aware multi-horizon system capable of predicting daily traffic and toll revenue for the next day, next week and next month.